

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas

Master of Science in Computer Engineering

Aug 2024 – May 2026

Fayetteville, AR

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Bachelor of Engineering in Electronics and Communication

Aug 2019 – May 2023

India

TECHNICAL SKILLS

Data Engineering: Flink SQL, Apache Kafka, Apache Flink, Apache Airflow, ClickHouse, MySQL, PostgreSQL

Cloud and Tools: AWS Glue, AWS Lambda, REST APIs, Test automation, Data validation and quality, ES6+, HTML5, CSS3, Amazon S3, Git, Grafana, HTML, CSS, Amazon EC2, Amazon RDS, CI/CD, TDD

Machine Learning and AI: LangChain, RAG, PyTorch, Scikit-learn, Flask, Pandas, FAISS

Languages: SQL, Python, C#

PROFESSIONAL EXPERIENCE

Data Engineering Intern

InsOps Inc.

Jul 2024 – Present

Chicago, Illinois

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant

University of Arkansas, Dept. of Computer Science and Engineering

Sep 2024 – May 2026

Fayetteville, AR

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant

AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Fayetteville, AR

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer

Soliton Technologies

Jun 2023 – Jun 2024

India

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements

PROJECTS

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama, Llama 3.2

- Built an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, chunk, and preprocess unstructured web content on demand
- Designed a local vector store with FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion and reducing processing latency
- Implemented a RAG pipeline via LangChain orchestrating semantic queries against vectorized data using local Llama 3.2 instances

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, Flink SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to handle out-of-order streams
- Designed ClickHouse MergeTree schemas sorted by composite keys to align disk layout with clinical query patterns, with an optimized Python multi-topic batch consumer isolating ingestion